Week 5 Notes & Quiz

Notes for all five Week 5 lectures, made only from the 131 slides in *WEEK 5.pdf*. Each yellow tag such as S13 is the slide (PDF page) the point comes from. The 50-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L21	**Fundamentals of Supply Chain Management (SCM)**	S1–S23
L22	**SC Operations: Manufacturing versus Service SC**	S24–S48
L23	**Principle Streams of Supply Chain: sourcing, procurement, supply**	S49–S67
L24	**Principle Streams of Supply Chain: operations, materials, production, logistics, distribution**	S68–S100
L25	**Objectives of Supply Chain Networks**	S101–S131
+	**Cheat sheet · Slide quirks · Quiz (50 Qs)**	—

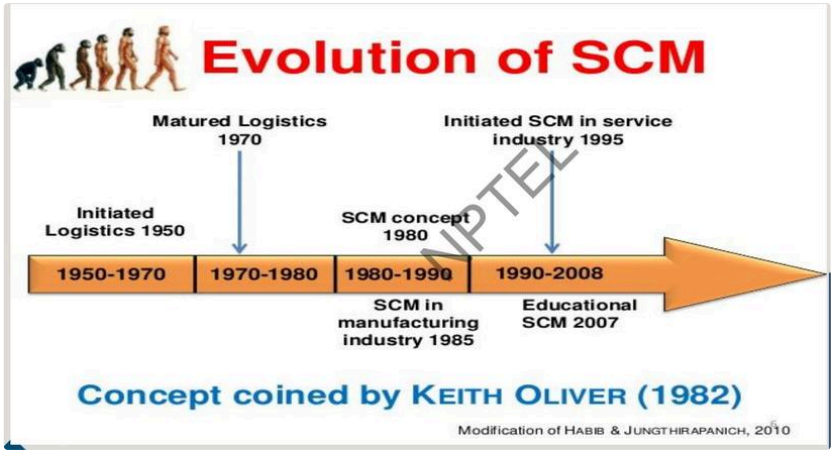
LECTURE 21 S1–S23

Fundamentals of Supply Chain Management

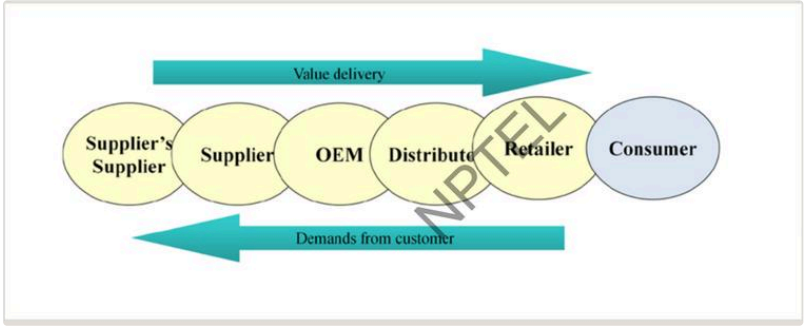
- Example chain (detergent): timber → paper → packaging; oil company → chemicals and plastics → P&G → supermarket or 3rd-party DC → Big Bazaar → customer. S3

> SUPPLY CHAIN · CHOPRA & MEINDL (2013)
>
> *A supply chain consists of all parties involved, directly or indirectly, in fulfilling a customer request. It includes not only the manufacturer and suppliers, but also transporters, warehouses, retailers, and even customers themselves.* S5
>
> Within a firm it includes new product development, marketing, operations, distribution, finance and customer service. Dawei & Chen: a group of interconnected companies that add value to inputs from source to end product. S5

- SCM combines **demand and supply networks**; each link tries to cut cost and improve long-run performance. S6



Original slide S7. Logistics initiated 1950 → matured 1970 → SCM concept 1980 → SCM in manufacturing 1985 → SCM in services 1995 → educational SCM 2007. Concept coined by **Keith Oliver (1982)**.



Original slide S8, basic SC model: supplier's supplier → supplier → OEM → distributor → retailer → consumer. Value moves downstream; demand moves upstream.

- Three **flows** between manufacturer and customer: **information, product, funds**. S9
- SCM network stakeholders: suppliers, manufacturers, warehouses and DCs, customers; costs at each link: material, transportation, manufacturing, inventory. S10
- Objective: maximize the overall value generated. Listed aims include timely delivery, effective inventory, flexibility, cost-effective production, collaboration, optimized delivery, customer value, quality, integrated technology. S11

Supply chain decisions (three phases)

PHASE	HORIZON	EXAMPLES
1. Strategy or design	Several years	Outsource or in-house; location and capacity of plants and warehouses; products at each location; transport modes; information system. S12
2. Planning	**A quarter to a year**	Configuration fixed; maximize **supply chain surplus**. Which markets from which locations, inventory policies, subcontracting, backup locations. S13
3. Operation	**Weekly or daily**	Individual orders: set due dates, pick lists, allocate orders, delivery schedules, replenishment orders. S14

Types of supply chain models 6 items

Continuous flow	Traditional; mature industries; same product, steady demand. S15
Agile	Unpredictable demand or custom orders; prioritizes flexibility. S15
Fast	Short-life-cycle products; ride a trend and sell out before it ends. S15
Flexible	**Seasonal** businesses; ramp up or wind down. S16
Efficient	Very tight profit margins; most efficient use of equipment and inventory. S16
Custom	Highly specialized industries, e.g. automobile makers. S16

Concept of SCM: three components

COMPONENT	ABOUT	DECISION LEVEL
Configuration	How the chain is built: supply base, extent of **vertical integration**, outsourcing, distribution channel. Also called **supply chain architecture**. S18	Strategic
Relationship	Inter-firm ties, mainly OEM with first-tier suppliers and customers. Only volume and price exchanged = **arm's length**; vision, investment and NPI plans shared = **close partnership**. S19	Strategic and operational
Coordination	Inter-firm operational coordination of material flows; inventory is a key focus; capacity, forecasting, scheduling. S20	Operational

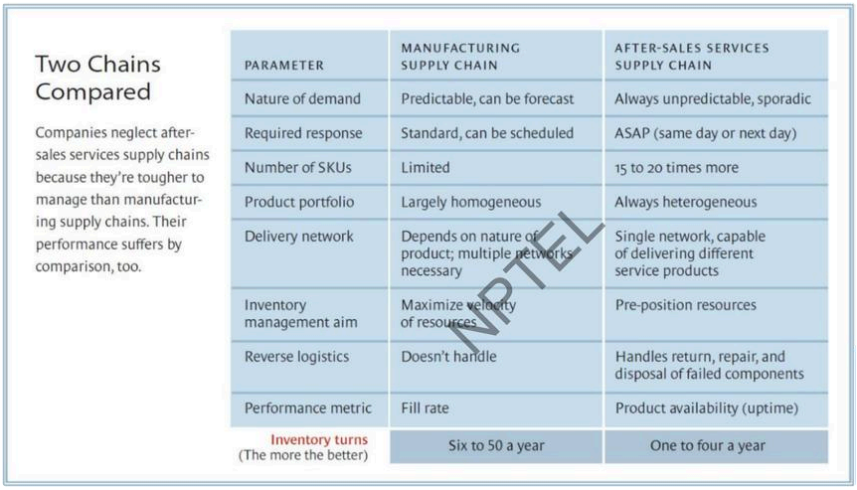
LECTURE 22 S24–S48

SC Operations: Manufacturing versus Service SC

- SC operations: supplier sourcing, supplier-purchaser collaboration, supplier information management, purchase order management, inventory, warehouse and transport management, SC planning, shipping. S26
- Four functions of operations management: **planning, scheduling, control, improvement**. S27
- SCM is described as a **subset of operations management**, managing material and information flow among suppliers and customers. S28
- Five unique attributes of services: **intangibility, perishability, heterogeneity, inseparability, customer participation**. Perishability mirrors inventory management; heterogeneity etc. have manufacturing counterparts like modularity and JIT. S29
- Manufacturing SC: flow of raw materials, components and information through production, assembly, packaging and distribution. Example model: three-layer supplier–manufacturer–retailer model with inspection for defective items (Pal & Mahapatra, 2017). S31–S32
- Service SC: delivering intangible services (healthcare, transport, hospitality, professional services); customer interaction is central. Supplier → service organization (supplier relationship, operation, customer relationship) → customer. S33–S34
- Service SC framework: strategy balances **competitive service requirements with cost and investments**; Plan, Source, Deliver, Return (repair and recovery, disposition). S35

CHARACTERISTIC	MANUFACTURING	SERVICE
SC input	Suppliers' materials and forecast information	Customer participation (presence, product, information)
Value-added process	Physical production and intangible services	Tangible and intangible services
Demand uncertainty management	**Inventory**	**Queues**
SC output	Pre-defined spec, made unique through **modularity**	Heterogeneous service
Production system	**Push** (sell from inventory)	**Pull** (initiated by customer demand)
Labour intensity · customer involvement · heterogeneity	Low	High
Measuring quality	Easy	Hard
Inventory	Exists	Does not exist

From S38 and S40.



Original slide S39 (HBR, "Winning in the Aftermarket"). After-sales chains: unpredictable demand, ASAP response, **15–20 times more SKUs**, measured by uptime, **1–4 inventory turns a year** vs 6–50 for manufacturing.

- Sector examples: manufacturing = automotive, food processing, make-to-order engineering, electronics, heavy engineering, pulp and paper, steel; service = regional hospital, printing and scanning, 3PL, heat treatment, regional utility, hardware retail. S43
- **Amazon case**: Seattle; started buying books from a distributor per order; in 2009 about **20 US and 30 overseas warehouses**; outbound shipping cost almost **$2 billion** (2009); 460,000+ Kindle books. Bought **Zappos** (2009: 121,000 product descriptions, 2.2 million images; added variety) and **diapers.com** (2010: little variety, big volume). S44–S45

LECTURE 23 S49–S67

Principle Streams of SC: Sourcing, Procurement, Supply

STREAM	COVERS S51–S53
1. Procurement, supply & sourcing	Acquiring goods and services: suppliers, contracts, relationships, timely quality delivery.
2. Operations, material & production	Converting inputs to products or services: inventory, planning, scheduling, quality.
3. Logistics & distribution	Movement and storage from production to consumption.

Sourcing management

- Systematic identification, assessment and enlistment of suppliers, usually via **tendering**; tactical and strategic. Usually the **first step** in the supply chain process; balances quality with affordability. S55

TYPE OF SOURCING	MEANING S56–S58
Outsourcing	An outside party does work previously done in-house.
Insourcing	Delegate to a team within the company; saves cost, quality monitored on the ground.
Near-sourcing	Operations placed close to where end products are sold.
Low-cost country sourcing (LCCS)	Countries with low labour costs; **China** is the go-to country.
Global sourcing	Buying across geopolitical boundaries.
Prime/subcontracting	A subcontractor does part of a larger project.
Captive service operations	Own entity overseas: more control and confidentiality, but **economies of scale suffer**.
Vertical integration	Buy your input supplier = **backward**; buy distribution = **forward** integration.
Few or many suppliers	Multi-supplier for commodities (price-based) vs single-source.
Virtual enterprise	Independent firms linked by IT to share skills, costs and markets.
Others	Professional services, manufacturing, joint ventures.

Procurement management

- Overseeing and administering all methods of acquiring materials, products and services; needs continuous monitoring, not one-off. S59
- Fundamentals: understand your purchasing portfolio, organize purchases, control inventory, include procurement in product development, risk management and KPIs, customer needs, look outside your market, prioritize relationships, collect spend data, know your negotiating position. S60
- Example: a construction company sourcing cement and steel for a building project. S62

8 steps of procurement S61

1. Identifying and planning
2. Choosing and selecting suppliers
3. Negotiating the contract
4. Placing the first purchase order
5. Accelerating the order
6. Receipt and inspection
7. Clearing the invoice and payment
8. Maintaining relationships and records

PROCUREMENT	SOURCING S63
Procures goods and services the firm needs	Finds, vets and contracts suppliers
Focuses on the **what** of supplies	Focuses on the **who**
Runs already-created supply chains	Builds and manages supply chains; builds alternatives for resilience

Supply management

- Identifying, acquiring and managing resources and suppliers essential to operations. Goals: **cost control, efficient resource allocation, risk management, gathering information** for decisions. S64
- Role: source and procure per leaders' wishes; build and hold suppliers accountable; use technology; consider supply and demand theory. S65

LECTURE 24 S68–S100

Principle Streams: Operations, Materials, Production, Logistics, Distribution

Operations management

- Optimizing operations to maximize efficiency, productivity and profit; converting inputs (materials, labour, equipment, technology) into outputs. S71
- Managers handle **process design**, balance resources and costs, shape product design, oversee strategy, streamline workflows. S72
- Key concepts: **processes, capacity, utilisation, flow**. Process: planning → scheduling → execution. S73–S74
- Types of operations: manufacturing, service, retail, hospitality, logistics, supply chain. S75–S76
- Strategies: reduce costs, improve quality, productivity, use data, set goals and forecast, collaboration, HRM, improve supply chain. S77

Materials management

- Planning, organizing and controlling the flow of materials a manufacturer needs: from **material requirement planning** to sourcing, storage, control and distribution. Goal: right quantity, right time, right cost. S78
- Primary objectives: low prices, high inventory turnover, low procurement cost, continuous supply, consistent quality, low payroll cost, good supplier relations. Secondary: reciprocal relations, new materials, make-or-buy, standardization, product improvement, interdepartmental harmony, forecasts. S79
- Components: **MRP** (quantity and timing of raw materials), purchase planning (balance holding vs ordering cost), inventory control, material supply management, quality control. S80–S81
- Significance: cost reduction, quality, timely delivery, inventory optimization, efficient supply chain. Challenges: SCM complexity, cost control, quality control, data management. S82 S84
- Improves profitability via procurement, logistics, inventory carrying, waste and manufacturing costs, and lower working capital. S85

Production management

- **Elwood S. Buffa**: decisions on the production process to make goods to specification, in the right amount, at the right time, within budget. S86
- **E.L. Brech**: "the process of effective planning and regulating the operations of that section of an enterprise which is responsible for the actual transformation of materials into finished products" (criticized for ignoring the human factor). S86
- Uses the 6 M's: Man, Machine, Methods, Money, Materials, Markets. S86
- Types: **intermittent** (project, job shop, batch) and **continuous** (mass and flow, process). S87
- Objectives: right quantity, quality, time and cost. Functions: product and resource selection, production planning and control, machine maintenance and replacement. S88–S89
- Strategies: continuous improvement, JIT, **TPM** (total productive maintenance), SCM. S90

Logistics and distribution management

- Logistics management: planning, implementing and controlling movement and storage of goods, services and information; right product, place, time and condition at minimum cost. S93
- Benefits: higher profitability, better delivery productivity, lower operating expenses, customer satisfaction, better intermodal operations, advanced route planning. S94
- Components of logistics management systems: **WMS, TMS, DMS**. TMS acts as a mediator between legacy order processing and warehouse distribution. S95–S96
- Distribution management systems: automated order management, inventory allocation, warehouse forecasting. S97
- Challenges in distribution: insufficient skilled resources, inadequate capital, disconnected technical infrastructure, multi-brand distributors, reluctance to modernize. S98

LECTURE 25 S101–S131

Objectives of Supply Chain Networks

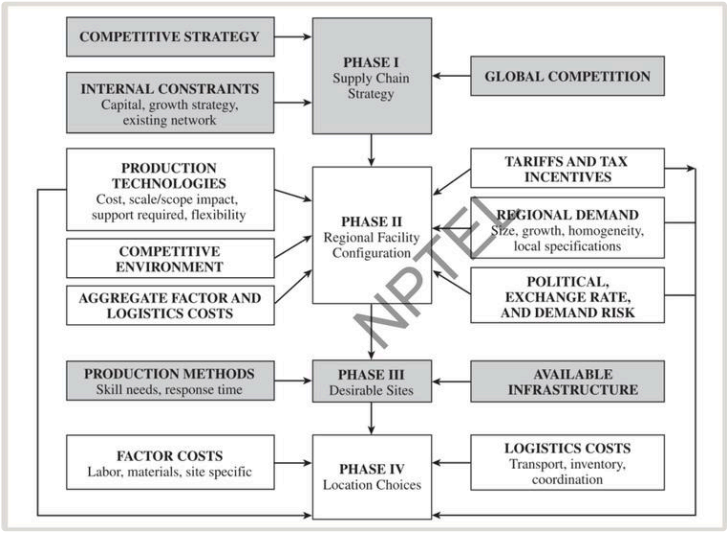
- Objective: maximize overall value generated. **Supply chain profitability** = what the customer pays minus the firm's cost to supply it; the higher it is, the more value generated. The only source of revenue is the **customer**. S103–S104
- Other objectives: cost of quality, replenishment, speedy delivery, shorter order time, efficiency, delivery optimization, awareness, class performance, inventory optimization, demand fulfilment, flexibility, business characteristics, efficacy, cost reduction, better decisions, better distribution. S104–S109

Supply chain network design

- A supply chain network is the interconnected system of organizations, facilities, activities and resources that produce and deliver goods or services. S111
- CIPS: network design "covers all of the movements and storage of raw materials, work-in-process inventory, and finished goods from the point-of-origin to the point-of-consumption." S111
- Design decisions: **facility role**, **facility location**, **capacity allocation**, **market and supply allocation**. S113

Factors influencing network decisions 8 items

Strategic	Cost leaders pick lowest-cost locations (**Foxconn, Flextronics** in China); responsive firms locate near markets (**Zara** in Portugal and Spain). S115
Technological	Big economies of scale → few high-capacity plants (**semiconductors**); low fixed cost → many local plants (**Coca-Cola bottling**). S116
Macroeconomic	Taxes, tariffs, exchange rates, shipping costs. S117
Infrastructure	1990s firms chose Shanghai, Tianjin, Guangzhou for infrastructure, not lowest cost. S117
Political	Stability; **Global Political Risk Index (GPRI)** by **Eurasia Group**: government, society, security, economy. S118
Competitive	Locate near or far from competitors. S119
Customer response time	Convenience stores: many, close to customers; supermarkets: fewer, larger. S120
Logistics and facility costs	More facilities → **inventory and facility costs rise**, **transport costs fall** until inbound economies of scale are lost. S121



Original slide S122 (Chopra & Meindl). Phase I supply chain strategy → Phase II regional facility configuration → Phase III desirable sites → Phase IV location choices, with the inputs feeding each phase.

Phase I · SC strategy	Broad design: stages, in-house vs outsourced, from the competitive strategy; forecast global competition; capital constraints. S123
Phase II · Regional configuration	Regions, roles, approximate capacity. **Homogeneous** needs → large consolidated facilities; varied needs → smaller local ones. Big economies of scale → few plants (e.g. AMD). S124
Phase III · Desirable sites	**Hard infrastructure**: suppliers, transport, communication, utilities, warehousing. **Soft infrastructure**: skilled workforce, turnover, community receptivity. S125
Phase IV · Location choices	Precise location and capacity for each facility to **maximize total profit**. S126

Case: Zara (Inditex)

- 2009: about **€11 billion** sales, 4,700+ outlets in about 76 countries. Design-to-sales cycle **4–6 weeks** vs 6+ months for the industry; new designs every week; 75% of display changed every 3–4 weeks; about **half the markdowns** of competitors. S127
- Flexible sources in Europe (Portugal, Spain) and low-cost sources in Asia; Inditex owns about **40%** of capacity. **Uncertain-demand** products from Europe, predictable ones from Asia. Over 40% of purchases after the season starts (vs under 20% typically). S128
- **Eight DCs in Spain**; delivery within **24 hours** in Europe, up to **48 hours** to America or Asia; several shipments a week. S129

BEFORE THE EXAM ALL LECTURES

Cheat sheet

3 flows	Information, product, funds S9
3 decision phases	Strategy (years) · planning (quarter–year) · operation (weekly/daily) S12–S14
6 SC models	Continuous flow, agile, fast, flexible, efficient, custom S15–S16
3 SCM concepts	Configuration (strategic) · relationship (both) · coordination (operational) S17
5 service attributes	Intangibility, perishability, heterogeneity, inseparability, customer participation S29
3 streams	Procurement/supply/sourcing · operations/material/production · logistics/distribution S51
6 M's	Man, Machine, Methods, Money, Materials, Markets S86
4 network decisions	Facility role, location, capacity allocation, market and supply allocation S113
4 design phases	SC strategy → regional configuration → desirable sites → location choices S122

NUMBER	MEANING	SLIDE
1982	Keith Oliver coins SCM	S7
15–20×	More SKUs in after-sales chains	S39
6–50 vs 1–4	Inventory turns: manufacturing vs after-sales	S39
$2 bn	Amazon outbound shipping cost, 2009	S44
4–6 weeks	Zara design-to-sales cycle	S127
8 DCs · 24 h / 48 h	Zara: DCs in Spain · delivery Europe / Americas-Asia	S129

MEMORY HOOK (MY OWN, NOT FROM THE SLIDES)

Phases of network design go from wide to narrow: **Strategy → Region → Sites → Spot**.

READ ONCE SOURCE CHECK

Slide quirks to watch

- Supply chain coordination is said to work "in a preferably **JIM** manner"; read it as JIT. S20
- The definition credits "Daeai & Chen (2013)", while the reference list has Lu Dawei & Chen (2011). S5 S22
- In the push/pull table, the "Suppliers" and "Responsiveness" rows are merged: "not critical" vs "must be responsive" belongs to responsiveness, and "often at a low level" vs "critical to overall success" to customer relations. S40
- Phase IV on S126 repeats a sentence from Phase I ("The supply chain strategy then specifies…").
- Video or link-only slides: S4, S30, S110. S41, S42 continue the comparison table (nature of output, inventory, lead times, customer engagement, quality, demand variability).

SELF-TEST 50 QUESTIONS

Week 5 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 21

According to Chopra & Meindl, a supply chain consists of:

- **A** all parties involved, directly or indirectly, in fulfilling a customer request
- **B** only retailers and customers
- **C** only the logistics department
- **D** only manufacturers and suppliers

Q2 LECTURE 21

The three flows in a supply chain are:

- **A** material, machine, man
- **B** cash, credit, debt
- **C** information, product, funds
- **D** product, people, power

Q3 LECTURE 21

Per the 'Evolution of SCM' slide, the SCM concept was coined by Keith Oliver in:

- **A** 1982
- **B** 1970
- **C** 1995
- **D** 2007

Q4 LECTURE 21

The objective of every supply chain should be to:

- **A** minimize inventory
- **B** maximize the overall value generated
- **C** maximize the number of suppliers
- **D** minimize transport only

Q5 LECTURE 21

Supply chain planning decisions typically cover a time frame of:

- **A** a week
- **B** a quarter to a year
- **C** a single day
- **D** several years

Q6 LECTURE 21

Setting order due dates and generating pick lists are decisions in which phase?

- **A** Supply chain strategy
- **B** Supply chain operation
- **C** Supply chain planning
- **D** Network design

Q7 LECTURE 21

Which supply chain model suits companies affected by seasonality?

- **A** Flexible
- **B** Fast
- **C** Continuous flow
- **D** Efficient

Q8 LECTURE 21

Which model aims to capitalize on a trend with short-life-cycle products?

- **A** Continuous flow
- **B** Custom
- **C** Agile
- **D** Fast

Q9 LECTURE 21

Supply chain configuration is also known as:

- **A** supply chain coordination
- **B** supply chain architecture
- **C** supply chain surplus
- **D** arm's length relationship

Q10 LECTURE 21

A relationship in which firms exchange only volume and price is called:

- **A** arm's length
- **B** vertical integration
- **C** close partnership
- **D** joint venture

Q11 LECTURE 22

Which are the four main functions of operations management in the supply chain?

- **A** Sort, set, shine, sustain
- **B** Define, measure, analyze, control
- **C** Planning, scheduling, control, improvement
- **D** Plan, source, make, deliver

Q12 LECTURE 22

Which is NOT one of the five unique attributes of services?

- **A** Intangibility
- **B** Customer participation
- **C** Perishability
- **D** Modularity

Q13 LECTURE 22

In service supply chains, demand uncertainty is managed through ____, while manufacturing uses ____.

- **A** inventory; queues
- **B** queues; inventory
- **C** forecasts; queues
- **D** modularity; inventory

Q14 LECTURE 22

The production system in a services supply chain is typically:

- **A** pull (initiated by customer demand)
- **B** push (sell from inventory)
- **C** batch
- **D** continuous

Q15 LECTURE 22

Measuring quality in a services supply chain is described as:

- **A** automated
- **B** unnecessary
- **C** hard
- **D** easy

Q16 LECTURE 22

In the HBR 'Two Chains Compared' table, after-sales service chains have about how many more SKUs?

- **A** 2–3 times
- **B** 15–20 times
- **C** 5 times
- **D** 100 times

Q17 LECTURE 22

Inventory turns in after-sales service supply chains are about:

- **A** daily
- **B** one to four a year
- **C** six to fifty a year
- **D** fifty to a hundred a year

Q18 LECTURE 22

The performance metric for after-sales service supply chains is:

- **A** inventory turns
- **B** on-time shipping
- **C** product availability (uptime)
- **D** fill rate

Q19 LECTURE 22

Which Amazon acquisition 'added little variety but considerable shipping volumes'?

- **A** Kindle
- **B** Zappos
- **C** diapers.com
- **D** Whole Foods

Q20 LECTURE 22

Amazon's outbound shipping-related costs in 2009 were almost:

- **A** $200 million
- **B** $2 billion
- **C** $20 billion
- **D** $460 million

Q21 LECTURE 23

Which are the three principle streams of supply chain?

- **A** Procurement/supply/sourcing; operations/material/production; logistics/distribution
- **B** Inbound, outbound, reverse
- **C** Finance, marketing, HR
- **D** Plan, make, deliver

Q22 LECTURE 23

In most organizations, the first step in the supply chain process is:

- **A** returns
- **B** production
- **C** distribution
- **D** sourcing

Q23 LECTURE 23

Placing some operations close to where end products are sold is called:

- **A** near-sourcing
- **B** captive service operations
- **C** low-cost country sourcing
- **D** insourcing

Q24 LECTURE 23

Which country is named as the go-to destination for low-cost country sourcing?

- **A** China
- **B** Mexico
- **C** India
- **D** Vietnam

Q25 LECTURE 23

When a company acquires its input supplier, it is called:

- **A** backward integration
- **B** horizontal integration
- **C** virtual enterprise
- **D** forward integration

Q26 LECTURE 23

Captive service operations give more control, but what is the drawback mentioned?

- **A** No control over quality
- **B** Economies of scale are negatively affected
- **C** Higher tariffs
- **D** Loss of confidentiality

Q27 LECTURE 23

What is step 5 of the 8 steps of procurement management?

- **A** Negotiating the contract
- **B** Receipt and inspection
- **C** Clearing the invoice
- **D** Accelerating the order

Q28 LECTURE 23

Procurement focuses on the ____ of supplies, while sourcing focuses on the ____.

- **A** what; who
- **B** who; what
- **C** cost; time
- **D** quality; quantity

Q29 LECTURE 23

A network of independent companies linked by information technology to share skills, costs and markets is a:

- **A** prime contractor
- **B** captive operation
- **C** joint venture
- **D** virtual enterprise

Q30 LECTURE 23

Which is NOT listed as a main goal of supply management?

- **A** Risk management
- **B** Cost control
- **C** Maximizing advertising spend
- **D** Efficient allocation of resources

Q31 LECTURE 24

The four key concepts of operations management on the slide are:

- **A** man, machine, money, materials
- **B** processes, capacity, utilisation, flow
- **C** cost, quality, time, flexibility
- **D** plan, do, check, act

Q32 LECTURE 24

MRP in materials management stands for:

- **A** Material Requirements Planning
- **B** Maximum Retail Price
- **C** Material Return Policy
- **D** Manufacturing Resource Process

Q33 LECTURE 24

Purchase planning aims to balance:

- **A** inventory holding costs and ordering costs
- **B** labour and machine costs
- **C** sales and marketing costs
- **D** tax and tariff costs

Q34 LECTURE 24

Production management attempts to use the 6 M's. Which is NOT one of them?

- **A** Money
- **B** Markets
- **C** Management
- **D** Methods

Q35 LECTURE 24

Whose definition of production management is criticized for ignoring the human factor?

- **A** Keith Oliver
- **B** Elwood S. Buffa
- **C** Peter Drucker
- **D** E.L. Brech

Q36 LECTURE 24

Job shop production is a type of:

- **A** intermittent production
- **B** continuous production
- **C** process production
- **D** mass production

Q37 LECTURE 24

TPM, a production management strategy, stands for:

- **A** Total Production Management
- **B** Total Productive Maintenance
- **C** Transport Planning Method
- **D** Time Per Machine

Q38 LECTURE 24

The three components of logistics management systems are:

- **A** ERP, CRM, SCM
- **B** EDI, GTM, RFID
- **C** MRP, JIT, TQM
- **D** WMS, TMS, DMS

Q39 LECTURE 24

Which system works as a mediator between legacy order processing and warehouse distribution?

- **A** TMS
- **B** DMS
- **C** MRP
- **D** WMS

Q40 LECTURE 24

Which is NOT listed among challenges in distribution management?

- **A** Excess skilled resources
- **B** Inadequate capital
- **C** Reluctance towards modernization
- **D** Multi-brand distributors

Q41 LECTURE 25

Supply chain profitability is the difference between:

- **A** revenue and tax
- **B** fixed and variable costs
- **C** what the customer pays and the firm's cost of supplying the product
- **D** inventory and sales

Q42 LECTURE 25

Which is NOT one of the supply chain network design decisions?

- **A** Facility location
- **B** Facility role
- **C** Advertising budget
- **D** Capacity allocation

Q43 LECTURE 25

Foxconn and Flextronics locating factories in China illustrates which factor?

- **A** Customer response time
- **B** Strategic (cost leadership)
- **C** Political
- **D** Infrastructure

Q44 LECTURE 25

Why does Coca-Cola set up many local bottling plants worldwide?

- **A** High economies of scale in bottling
- **B** Tax holidays
- **C** Political pressure
- **D** Low fixed costs of facilities, so many local plants cut transport cost

Q45 LECTURE 25

The Global Political Risk Index (GPRI) is evaluated by:

- **A** Gartner
- **B** Eurasia Group
- **C** World Bank
- **D** IMF

Q46 LECTURE 25

As the number of facilities in a supply chain increases:

- **A** inventory and facility costs decrease
- **B** transportation costs decrease (until inbound economies of scale are lost), while inventory and facility costs increase
- **C** all costs increase
- **D** all costs decrease

Q47 LECTURE 25

Which phase of network design identifies regions, facility roles and approximate capacity?

A Phase IV
B Phase III
C Phase I
D Phase II

Q48 LECTURE 25

Availability of a skilled workforce and community receptivity are examples of:

A macroeconomic factors
B soft infrastructure
C hard infrastructure
D factor costs

ANSWERS 50 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	A	It includes transporters, warehouses, retailers and even customers. S5
Q2	C	S9. S9
Q3	A	S7: 1982. Logistics initiated 1950, matured 1970; SCM in services 1995; educational SCM 2007. S7
Q4	B	S11. S11
Q5	B	Strategy: years; planning: quarter to a year; operation: weekly or daily. S13
Q6	B	S14: operation, weekly or daily. S14
Q7	A	S16: the flexible model ramps production up or down. S16
Q8	D	S15. S15
Q9	B	S18; it is a strategic, higher-level decision. S18
Q10	A	S19. Sharing vision, investment and NPI plans makes it a close partnership. S19
Q11	C	S27. S27
Q12	D	Attributes: intangibility, perishability, heterogeneity, inseparability, customer participation. Modularity is a manufacturing counterpart. S29
Q13	B	S38: manufacturing → inventory; service → queues. S38
Q14	A	S40. S40
Q15	C	Manufacturing: easy; services: hard. S40
Q16	B	S39: 15 to 20 times more SKUs. S39
Q17	B	Manufacturing: 6–50 a year; after-sales: 1–4 a year. S39
Q18	C	Manufacturing chains use fill rate. S39
Q19	C	diapers.com (2010). Zappos (2009) added a lot of variety. S45
Q20	B	S44. S44
Q21	A	S51. S51
Q22	D	S55: sourcing, typically through tendering. S55
Q23	A	S56. S56
Q24	A	S56. S56
Q25	A	Backward = acquiring input supplier; forward = acquiring distribution. S58
Q26	B	S57. S57
Q27	D	1 identify/plan, 2 select suppliers, 3 negotiate, 4 first PO, 5 accelerate order, 6 receipt/inspection, 7 invoice/payment, 8 relationships/records. S61
Q28	A	S63. S63
Q29	D	S58. S58
Q30	C	Goals: cost control, efficient resource allocation, risk management, gathering information for decisions. S64
Q31	B	S73. S73
Q32	A	S80: determines quantity and timing of raw materials. S80
Q33	A	S80. S80
Q34	C	The 6 M's: Man, Machine, Methods, Money, Materials, Markets. S86
Q35	D	S86: E.L. Brech. S86
Q36	A	Intermittent: project, job shop, batch. Continuous: mass/flow, process. S87
Q37	B	S90: proactive maintenance to raise equipment efficiency. S90
Q38	D	Warehouse, Transportation and Distribution Management Systems. S95
Q39	A	S96: the Transportation Management System. S96
Q40	A	Challenges include insufficient (not excess) skilled resources. S98
Q41	C	S104: higher SC profitability means more value generated. S104
Q42	C	Facility role, location, capacity allocation, market and supply allocation. S113
Q43	B	S115: cost-leadership firms pick the lowest-cost locations. S115
Q44	D	Technological factor: low fixed cost favours many local facilities; semiconductors are the opposite. S116
Q45	B	Four categories: government, society, security, economy. S118
Q46	B	S121. S121
Q47	D	Phase II: regional facility configuration. S124
Q48	B	Hard: suppliers, transport, communication, utilities, warehousing. S125
Q49	A	The industry average is over six months. S127

Q	ANS	WHY · SLIDE
Q50	B	Predictable products come from Asia. S128

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